

DIY Private UK VPN

Business Analyst Project Portfolio

Irene Tsang · Personal Project · 2026

Google Cloud · Outline VPN · Kanban · Docker

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1. Business Requirements Analysis (BRA)

1.1 Project Overview

Project title: DIY Private UK VPN on Google Cloud

Project type: Personal infrastructure project

Project owner: Irene

Duration: May 2026

Methodology: Kanban (solo, flow-based delivery)

1.2 Problem Statement

As a British national and UK property owner residing in Guangzhou, China, I required a reliable UK IP address to access UK payment portals and government services. Commercial VPN providers were rejected due to:

- Unreliable UK IP assignment — not guaranteed to route through London
- High monthly cost (\$5–\$13/month) relative to usage (under 2 hours/month)
- Privacy concerns — shared infrastructure used by thousands of users
- Frequent blocking by the Great Firewall of China (GFW)
- Inability to customise obfuscation settings for GFW bypass

1.3 Business Objectives

Objective	Success Criteria
Establish a dedicated UK IP address	ipleak.net confirms IPv4 and IPv6 routing through London
Bypass the Great Firewall reliably	Connection succeeds from Guangzhou without interruption
Minimise monthly operating cost	Total cost under \$5/month for under 2 hours usage
Ensure full IPv4 and IPv6 coverage	No IP leaks detected on ipleak.net
Maintain privacy and sole ownership	No shared infrastructure — private server only

1.4 Stakeholders

Stakeholder	Role	Interest
Irene	Project owner and sole user	Reliable, low-cost UK IP for bill payment
UK entities	External system	Accepts payments from verified UK IP addresses

Stakeholder	Role	Interest
Google Cloud	Infrastructure provider	Compute and networking services
Outline / Jigsaw (Google)	Software provider	Open-source VPN software used for deployment

Engagement approach: as a solo project, Irene held Responsible and Accountable for every stakeholder decision. The remaining RACI roles — Consulted and Informed — were not applicable, since there was no team to delegate to.

1.5 Functional Requirements

ID	Requirement
FR-01	System must assign a static UK (London) IP address to all outbound traffic
FR-02	System must route both IPv4 and IPv6 traffic through the UK server
FR-03	System must obfuscate VPN traffic to bypass GFW deep packet inspection
FR-04	System must support multiple device connections via access keys
FR-05	System must allow VM to be stopped and restarted without IP address change
FR-06	System must provide a management interface (Outline Manager) for key administration

1.6 Non-Functional Requirements

ID	Requirement
NFR-01	Monthly cost must not exceed \$5 under normal usage patterns
NFR-02	Connection must be established within 60 seconds of VM start
NFR-03	System must remain operational without manual intervention once configured

ID	Requirement
NFR-04	Server must be located in the London (europe-west2) Google Cloud region
NFR-05	Solution must be maintainable by a non-technical user following documented steps

1.7 Constraints and Assumptions

Constraints

- The Great Firewall of China actively blocks standard VPN protocols — obfuscation is mandatory
- Google Cloud SSH browser tool is subject to GFW interference — a secondary VPN is required for server management
- Google Cloud free trial is limited to one account per payment method
- Static IP must be reserved separately to survive VM stop/start cycles

Assumptions

- User has access to a UK-issued payment card for Google Cloud billing
- User has a secondary VPN available (e.g. Psiphon) for initial server setup
- Ubuntu 22.04 LTS and Docker are sufficient for Outline VPN deployment
- e2-micro compute tier provides adequate performance for personal VPN use

1.8 Risks and Mitigations

Risk	Likelihood	Impact	Mitigation
GFW blocks Google Cloud IPs	Medium	High	Switch to Vultr or Hetzner London servers
VM IP changes on restart	High	Medium	Reserve static IP before installation
SSH connection drops during installation	High	Low	Re-run installer — idempotent script
Google Cloud account suspension	Low	High	Upgrade to paid account immediately
Outline Client conflicts with other VPN software	Medium	Medium	Uninstall conflicting VPN drivers before installation

1.9 Requirements Traceability

Each functional requirement is traced to the user stories that implement it and the evidence that verifies it:

Requirement	Traced To (User Stories)	Verification
FR-01	US-01, US-02, US-08	VM provisioned in europe-west2; static IP attached; ipleak.net confirms UK IPv4 address
FR-02	US-08	ipleak.net confirms UK IPv4 routing; IPv6 routing not independently verified by any acceptance criteria — flagged as a follow-up check
FR-03	US-04, US-07, US-08	Outline shadowbox container running; Client connects from Guangzhou; UK IP confirmed despite GFW deep packet inspection
FR-04	US-06, US-07	Outline Manager key list populated; per-device keys renamed and distributed via QR code
FR-05	US-02, US-09, US-11	Static IP persists across stop/start cycles; no reconfiguration required after restart
FR-06	US-05, US-06	apiUrl and certSha256 retrieved; Outline Manager dashboard shows green status

2. Product Backlog

2.1 Epic Overview

Epic ID	Epic Name	Description
EP-01	Infrastructure setup	Provision and configure cloud server in UK region
EP-02	VPN installation	Install and configure Outline VPN on the server
EP-03	Client connectivity	Connect devices to the VPN and verify UK IP
EP-04	Cost optimisation	Minimise monthly running cost for low-frequency usage
EP-05	Resilience and maintenance	Ensure system survives restarts, IP changes, and GFW interference

2.2 User Stories and Tasks

EP-01 — Infrastructure setup

Story ID	User Story	Acceptance Criteria
US-01	As a user, I want a cloud server in London so that my traffic appears to originate from the UK.	VM created in europe-west2 (London) region running Ubuntu 22.04 LTS
US-02	As a user, I want a static IP address so that my VPN configuration does not break when the server restarts.	Static external IP reserved and attached to VM; IP persists across stop/start cycles
US-03	As a user, I want firewall rules configured so that VPN traffic can reach my server.	TCP and UDP all ports open; Outline successfully receives incoming connections

Tasks for EP-01:

Task ID	Task	Status
T-01	Create Google Cloud account and enable billing	Done

Task ID	Task	Status
T-02	Create e2-micro VM in europe-west2 (London)	Done
T-03	Set boot disk to Ubuntu 22.04 LTS	Done
T-04	Create firewall rule — TCP + UDP all ports	Done
T-05	Reserve static external IP in europe-west2	Done
T-06	Attach static IP to VM network interface	Done

EP-02 — VPN installation

Story ID	User Story	Acceptance Criteria
US-04	As a user, I want Outline VPN installed on my server so that I can route traffic through a UK IP.	Outline shadowbox and watchtower Docker containers running; sudo docker ps shows both Up
US-05	As a user, I want the apiUrl string so that I can connect Outline Manager to my server.	sudo cat /opt/outline/access.txt returns valid certSha256 and apiUrl values

Tasks for EP-02:

Task ID	Task	Status
T-07	Connect to VM via SSH browser tool (with secondary VPN active)	Done
T-08	Run Outline install script via wget	Done
T-09	Confirm Docker installation (type Y when prompted)	Done
T-10	Resolve Watchtower container conflict (type Y to replace)	Done
T-11	Retrieve apiUrl from /opt/outline/access.txt	Done

Task ID	Task	Status
T-12	Verify static IP matches apiUrl IP — reinstall if mismatch	Done

EP-03 — Client connectivity

Story ID	User Story	Acceptance Criteria
US-06	As a user, I want Outline Manager connected to my server so that I can manage access keys.	Outline Manager shows green server dashboard with key list
US-07	As a user, I want Outline Client installed on my devices so that I can connect to the VPN.	Outline Client connects successfully and shows green status
US-08	As a user, I want to verify my UK IP so that I can confirm the VPN is working correctly.	ipleak.net shows UK IPv4 address with no Chinese IP leaks

Tasks for EP-03:

Task ID	Task	Status
T-13	Format apiUrl and certSha256 into JSON string	Done
T-14	Paste JSON string into Outline Manager — add server	Done
T-15	Rename default keys (Key 0, Key 1, Key 2) per device	Done
T-16	Share key to phone via QR code — install Outline Client	Done
T-17	Resolve NordVPN driver conflict on Windows laptop	Done
T-18	Verify UK IP on ipleak.net — confirm no leaks	Done

EP-04 — Cost optimisation

Story ID	User Story	Acceptance Criteria
US-09	As a user, I want to stop the VM when not in use so that I minimise my monthly cloud spend.	VM stopped after each use session; monthly cost confirmed below \$2.50 with static IP
US-10	As a user, I want a cost breakdown so that I can plan my monthly spend.	Cost model documented: ~\$0.44 storage + \$1.46 static IP = ~\$1.90/month when stopped

EP-05 — Resilience and maintenance

Story ID	User Story	Acceptance Criteria
US-11	As a user, I want the VPN to survive VM restarts so that I do not need to reconfigure devices after each session.	Static IP ensures Outline Client and Manager require no reconfiguration after restart
US-12	As a user, I want to know how to recover from SSH failures so that I can resolve issues independently.	Recovery procedure documented: retrieve apiUrl via access.txt; reinstall via idempotent script

3. Agile Methodology — Kanban

3.1 Why Kanban

Kanban was selected over Scrum for this project based on the following factors:

Factor	Rationale
Solo execution	Scrum ceremonies (standups, retrospectives, sprint planning) are designed for teams — Kanban suits individual contributors
Unpredictable blockers	GFW interference, SSH failures, and IP mismatches made sprint-based estimation impractical — work progressed as blockers were resolved
Continuous flow	Tasks were completed as soon as they were unblocked rather than batched into fixed sprints
Low overhead	Kanban's minimal process allowed focus on delivery rather than ceremony

3.2 Kanban Principles Applied

Principle	How it was applied
Visualise the workflow	Tasks tracked across Backlog → In Progress → Blocked → Done columns
Limit work in progress (WIP)	One epic addressed at a time — infrastructure before VPN before client
Manage flow	Blockers (GFW, SSH auth, IP mismatch) identified and resolved before progressing
Make policies explicit	Clear acceptance criteria defined per user story before work began
Improve continuously	Each failure (NordVPN conflict, suspended account, IP mismatch) fed directly into updated documentation

3.3 Kanban Board State — Final

	Items
Backlog	None — all items completed

	Items
In Progress	None — all items completed
Blocked	None — all blockers resolved
Done	T-01 through T-18 — all tasks complete

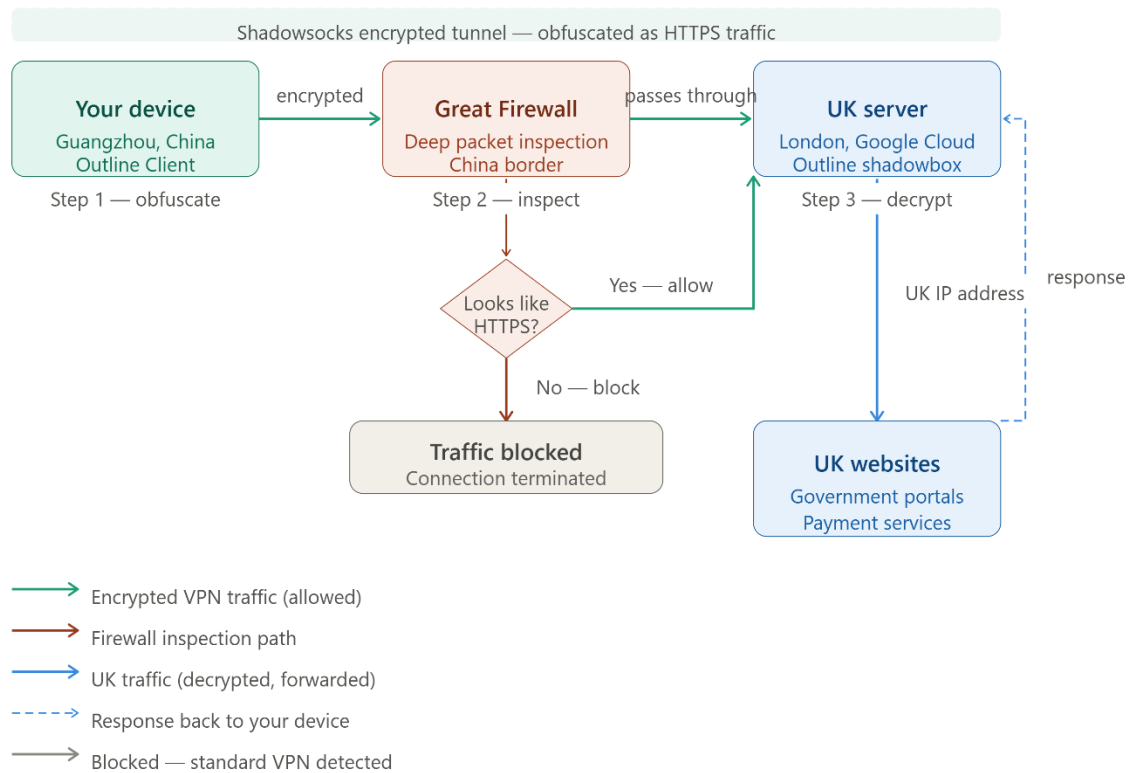
3.4 Key Blockers and Resolutions

Blocker	Root Cause	Resolution
SSH connection failing	GFW blocking Google Cloud SSH from China	Used secondary VPN (Psiphon) to establish SSH session
Outline Manager cannot connect	GFW blocking management port	Outline Manager requires secondary VPN; Outline Client does not
NordVPN driver conflict	Residual NordVPN TAP adapter blocking Outline's outline-tap0	Full uninstall of NordVPN + clean reinstall of Outline Client
Google Cloud account suspended	Free trial expired mid-project	Upgraded to paid account; VM and data preserved
SSH authentication failed	Account suspension invalidated SSH keys	Deleted old VM; created fresh VM with same static IP
Outline IP mismatch	Static IP not attached before Outline installation	Attached static IP first; reinstalled Outline to pick up correct IP

4. Process Flowchart — VPN Connection Routine

4.1 Monthly Bill Payment Flow

The following describes the end-to-end process for using the VPN to pay UK council bills from China. This represents the steady-state routine once the system is fully configured.



Step	Action	Actor	Notes
1	Open Google Cloud Console	User	Requires secondary VPN active in China
2	Navigate to Compute Engine → VM Instances	User	
3	Start VM — tick checkbox → click Start	User	VM takes ~60 seconds to fully boot
4	Wait for status dot to turn green	System	Do not proceed until VM shows green
5	Open Outline Client on device	User	Phone or laptop
6	Tap Connect	User	

Step	Action	Actor	Notes
7	Verify UK IP on ipleak.net	User	IPv4 must show London — abort if Chinese IP shown
8	Access UK payment portal	User	Portal accepts UK IP
9	Complete payment	User	
10	Disconnect Outline Client	User	Toggle off in app
11	Return to Google Cloud Console	User	
12	Stop VM — tick checkbox → click Stop	User	Reduces cost to ~\$0.50/month
13	Confirm VM status dot turns grey	System	Compute billing stops

4.2 Troubleshooting Decision Flow

If Outline Client fails to connect, follow this decision path:

Check	If Yes	If No
Is the VM running (green dot)?	Proceed to next check	Start the VM and wait 60 seconds
Does ipleak.net show a UK IP?	VPN is working — proceed	Check Outline Client is toggled on
Does Outline Client show an error?	Check error type below	Toggle off and on again
Is the error an IP mismatch?	SSH in — run reinstall script	Proceed to next check
Is another VPN running on the device?	Disconnect and quit it first	Reinstall Outline Client cleanly
Has the VM been recently restarted?	Wait 90 seconds and retry	Contact support or recreate VM

5. Retrospective and Lessons Learned

Looking back across the blockers logged in 3.4, a few things would change how I'd scope a similar project next time:

- NordVPN driver conflict — I would like to add “audit existing VPN clients on target devices” as an explicit task under EP-01, before installation, rather than discovering the conflict mid-connection. A five-minute check would have prevented a same-day support loop.
- Google Cloud account suspension — the free trial's billing-cycle risk was never captured as a requirement at all. In hindsight, “billing account must not lapse mid-build” belongs alongside NFR-03, since an unplanned outage is functionally the same failure whether it's technical or administrative.
- Outline IP mismatch — this only surfaced because installation ran before the static IP was attached. A dependency note on the EP-01/EP-02 tasks — “static IP must be attached before VPN installation” — would have caught it during planning instead of during troubleshooting.

More generally, treating each blocker as a documentation update worked well for capturing resolutions after the fact (see 3.2, “Improve continuously”). Still, a short pre-mortem against the constraints in 1.7 would have caught at least two of these three issues before they happened rather than after.

That's the gap between a backlog that tracks what got built and one that captures why it went the way it did — and it's the habit I'd bring into a BA role.

End of document